

The empirical recurrence for OEIS A250992: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A250992 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1249$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A250992 is “Number of $(n+1) \times (6+1)$ 0..2 arrays with nondecreasing $\min(x(i,j), x(i,j-1))$ in the i direction and nondecreasing absolute value of $x(i,j) - x(i-1,j)$ in the j direction.”. It has offset 1 and begins

28450, 255502, 1869400, 12340547, 77449279, 469255626, 2783159034, 16265322280, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:57:22, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2187$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 1249$ of the 2187, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1249$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 4374$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	95	c_{21}	1088544624305646886	c_{41}	3648621324841682491628902	c_{61}	−22
c_2	−4257	c_{22}	−2580365303709830882	c_{42}	−4399436658742238728631389	c_{62}	−162
c_3	119165	c_{23}	−2723349754096096710	c_{43}	−5761946541203009748525187	c_{63}	−16
c_4	−2323067	c_{24}	26810216449553851964	c_{44}	11756876715203662511760990	c_{64}	714
c_5	33207667	c_{25}	−38692696402029947068	c_{45}	6234650920925620396177208	c_{65}	219
c_6	−355389201	c_{26}	−104366358657757400094	c_{46}	−23561131230424036795154529	c_{66}	−21
c_7	2828235759	c_{27}	443379029047550358326	c_{47}	−2883365039174548209091793	c_{67}	−12
c_8	−15854810190	c_{28}	−160591580391545365164	c_{48}	38080003325598111944049454	c_{68}	31
c_9	49750997614	c_{29}	−2190463479978715983392	c_{49}	−4963781738256467226143048	c_{69}	43
c_{10}	67621109131	c_{30}	4155901706562696276130	c_{50}	−51545655838884781220136478	c_{70}	57
c_{11}	−1801940826115	c_{31}	4715196329444810096690	c_{51}	14979632583351134393915790	c_{71}	−7
c_{12}	10193543000854	c_{32}	−24943011401301920348761	c_{52}	59914825307293442865607170	c_{72}	−3
c_{13}	−22545154024800	c_{33}	10375839953178933257347	c_{53}	−22586071126475123264503584	c_{73}	−1
c_{14}	−70263345792913	c_{34}	86990476711848018567459	c_{54}	−60845840550321897925208332	c_{74}	38
c_{15}	723129182724079	c_{35}	−129074736405795119588351	c_{55}	23903018616282432482186320	c_{75}	13
c_{16}	−2139552867817292	c_{36}	−177609666229661114983835	c_{56}	54449358903331411428331128	c_{76}	2
c_{17}	−1382089948813506	c_{37}	577484995944175616199599	c_{57}	−18553935712053090376368624	c_{77}	2
c_{18}	32558042813655196	c_{38}	68981892807283229855087	c_{58}	−42780032092845658597573072	c_{78}	
c_{19}	−96418735457260492	c_{39}	−1701094094093038326294953	c_{59}	9857528844497666084029328		
c_{20}	−43468371489882516	c_{40}	985395577225806854616434	c_{60}	28956260494933140386463840		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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